

Homework B2 | ECON 0100 | Fall 2026 | Solutions

Solution guide for the teaching team. Answers and working are in red; everything in black is what students see. Students answer on paper and submit selections on Gradescope, where the homework is completion-graded.

Butterbeer

The supply curve for butter beer can be represented by the following equation:

$$P = 20 + \frac{1}{2}Q$$

Prices are in galleons and quantity is in bottles.

Q1 | Quantity Supplied

Use a graph to plot this supply curve, including the quantity supplied at both 40 galleons and 50 galleons.

- a) Quantity supplied at 40 galleons: 40 bottles
- b) Quantity supplied at 50 galleons: 60 bottles
- c) How much did quantity supplied change as the price dropped from 50 to 40? down by 20 bottles
- d) What is marginal cost at a quantity of 60? 50 galleons

The vertical intercept is 20 galleons: below that price nobody sells. Price to quantity: a) $40 = 20 + \frac{1}{2}Q$ gives $Q = 40$; b) $50 = 20 + \frac{1}{2}Q$ gives $Q = 60$; c) $40 - 60 = -20$, so quantity supplied fell by 20 bottles, the law of supply. Quantity to price: d) the height of supply at 60 bottles is $20 + \frac{1}{2} \cdot 60 = 50$ galleons, the marginal cost of the sixtieth bottle, which matches b).

Q2 | Producer Surplus

Then find and label the producer surplus at these prices.

- a) PS at 40 galleons: 400 galleons
- b) PS at 50 galleons: 900 galleons
- c) How much did producer surplus change as the price dropped from 50 to 40? down by 500 galleons

Producer surplus is the triangle above supply and below the price, out to the quantity sold; its height is the price minus the 20-galleon intercept.

a) $\frac{1}{2} \cdot 40 \cdot (40 - 20) = 400$ galleons. Revenue is the rectangle $40 \cdot 40 = 1600$; the 1200 under the curve covers marginal costs.

b) $\frac{1}{2} \cdot 60 \cdot (50 - 20) = 900$ galleons. Revenue is $50 \cdot 60 = 3000$.

c) $400 - 900 = -500$: producer surplus fell by 500 galleons, 10 less on each of the 40 bottles still sold (400) plus the triangle on the 20 bottles no longer sold (100).

